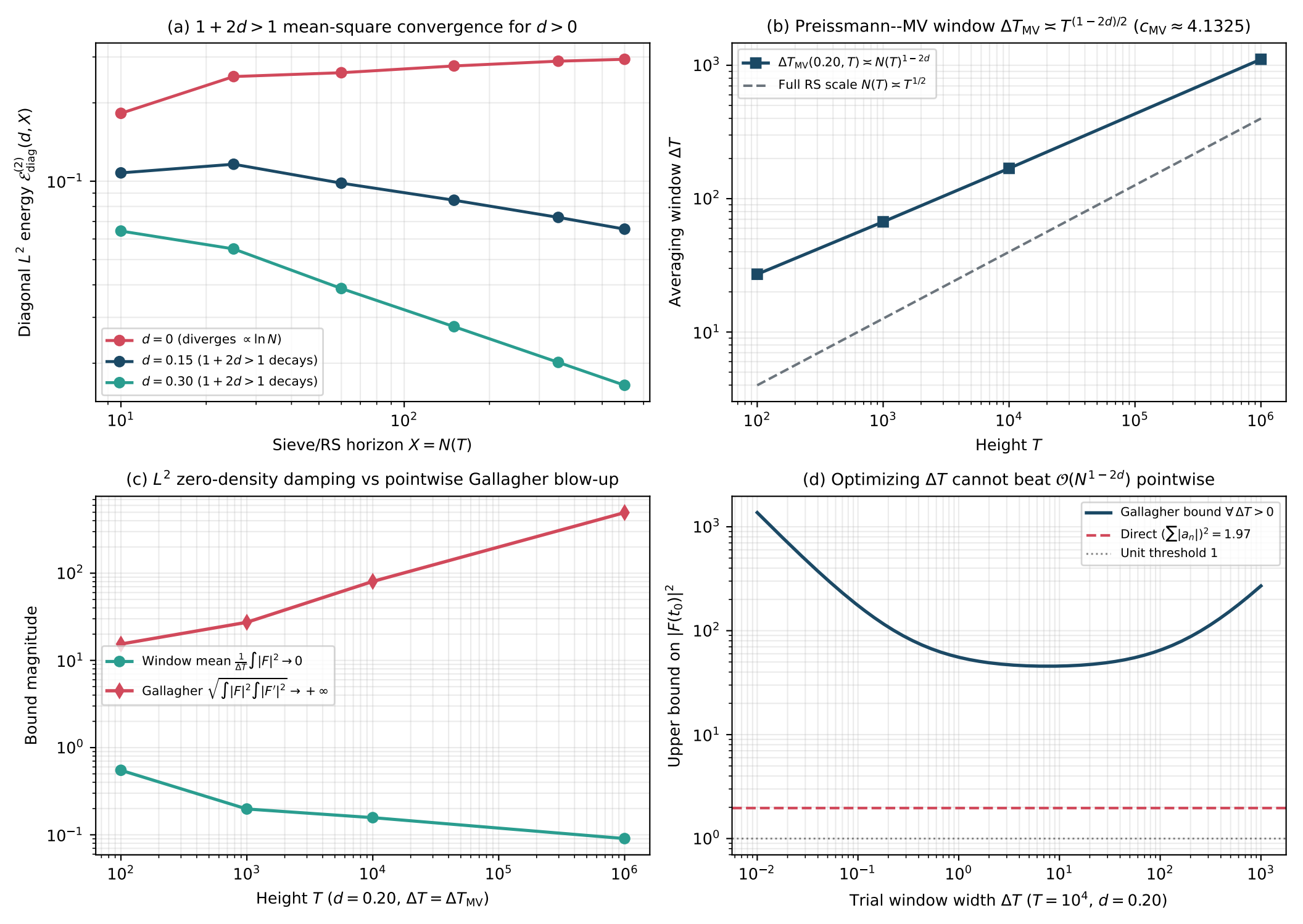
(a) $1+2d>1$ mean-square convergence for $d>0$
Diagonal L^2 energy $\mathcal{E}^{(2)}_{\mathrm{diag}}(d,X)$
10^{-1}
$d=0$ (diverges $\propto \ln N$)
$d=0.15$ ($1+2d>1$ decays)
$d=0.30$ ($1+2d>1$ decays)
10^1
10^2
Sieve/RS horizon $X=N(T)$
(b) Preissmann--MV window $\Delta T_{\mathrm{MV}} \asymp T^{(1-2d)/2}$ ($c_{\mathrm{MV}} \approx 4.1325$)
$\Delta T_{\mathrm{MV}}(0.20,T) \asymp N(T)^{1-2d}$
Full RS scale $N(T) \asymp T^{1/2}$
Averaging window ΔT
10^3
10^2
10^1
10^2
10^3
10^4
10^5
10^6
Height T
(c) L^2 zero-density damping vs pointwise Gallagher blow-up
Bound magnitude
10^2
10^1
10^0
10^{-1}
Window mean $\frac{1}{\Delta T}\int |F|^2 \to 0$
Gallagher $\sqrt{\int |F|^2 \int |F'|^2} \to +\infty$
10^2
10^3
10^4
10^5
10^6
Height T ($d=0.20$, $\Delta T=\Delta T_{\mathrm{MV}}$)
(d) Optimizing ΔT cannot beat $\mathcal{O}(N^{1-2d})$ pointwise
Upper bound on $|F(t_0)|^2$
Gallagher bound $\forall\, \Delta T>0$
Direct $(\sum |a_n|)^2 = 1.97$
Unit threshold 1
10^3
10^2
10^1
10^0
10^{-2}
10^{-1}
10^0
10^1
10^2
10^3
Trial window width ΔT ($T=10^4$, $d=0.20$)